

PRD - Onboarding Recovery & First-Use Activation

Hackathon | Purple Rockets | Group 1 | Onboarding | From sign-up through first transaction

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Executive summary

The onboarding funnel currently loses customers at multiple stages, with the largest drop occurring at selfie/liveness. Across 260,000 sign-ups, 115,192 customers complete onboarding (44.30%) and 50,024 activate within 30 days (19.24%). Of the 115,192 customers who complete onboarding, 65,168 do not make a first transaction within 30 days.

This PRD recommends two complementary MVPs

1. **Selfie Recovery — Version-Aware Update Nudge:** show an optional, pre-camera app-update nudge to Android customers on legacy versions, with a direct path to resume selfie/liveness.
2. **First-Use Activation Coach:** score customers who completed onboarding but have not yet transacted, place them into three propensity tiers, and select a constrained next-best action. Only the medium-propensity tier receives the higher-cost guided task with a pay-on-conversion incentive.

The product decision is to fix the onboarding experience and the post-onboarding activation gap, rather than change the acquisition-channel mix as the primary hackathon intervention. This gives the team one low-cost technical intervention and one differentiated AI-assisted intervention, while protecting customers who are already likely to convert.

Problem statement

Customers enter onboarding with different technical contexts, levels of intent and support needs, but the current journey does not sufficiently adapt to those differences.

We see two connected problems:

- **Onboarding friction:** the largest stage-level loss happens in selfie/liveness. Customers on Android and legacy app versions have substantially higher abandonment than customers on current Android versions.
- **Activation gap:** completing account creation is not enough. A large group finishes onboarding but never performs a first transaction, which means the product is not helping customers reach first value.

The business consequence is a compounding funnel loss. Customers who do not complete onboarding cannot activate, while customers who complete onboarding but do not transact represent a large opportunity that is not covered by the onboarding event stream.

Jobs to be done

During onboarding: "When I am blocked in identity verification, help me understand what to do next and let me resume without repeating unnecessary work."

After account creation: "When my account is ready, help me understand and complete the most relevant first action so I can experience value quickly."

Scope and non-goals

In scope

- Seven-step funnel reconstruction.
- Customer-level state and feature table.
- Selfie/liveness update-nudge prototype (Android only).
- Post-onboarding next-best-action prototype.
- Propensity-based targeting with deterministic guardrails.
- Constrained AI-generated copy.
- Experiment design and event taxonomy.

Out of scope for the hackathon

- Automated identity approval or rejection.
- Automated fraud decisions.
- Replacing compliance or customer support processes.
- Full production integration with app stores or operational queues.
- Changing acquisition budgets or channel mix without CAC and quality analysis.

- A causal claim that any observed correlation is the root cause of abandonment.

Current funnel discovery

The table below is calculated at customer level, using `steps_completed` (the step at which a non-completing customer's journey ends) joined against `completed_onboarding`. Retries generate multiple event rows in `onboarding_events`, so event-row counts must never be treated as customer counts — see §3.3.

Step	Stage	Customers entering	Completed	Abandoned	Stage abandonment	Cumulative reach	Current interpretation
1	Phone verification	260,000	247,309	12,691	4.88%	95.12%	Early friction, but not the main loss
2	Personal data	247,309	227,271	20,038	8.10%	87.41%	Form completion and data-entry friction
3	ID document upload	227,271	190,642	36,629	16.12%	73.32%	Document quality, capture and comprehension opportunity
4	Selfie / liveness	190,642	125,513	65,129	34.16%	48.27%	Largest stage-level drop; 44.98% of all observed abandonments
5	Address	125,513	118,672	6,841	5.45%	45.64%	Smaller but actionable form friction
6	Terms acceptance	118,672	115,192	3,480	2.93%	44.30%	Low-volume drop relative to earlier stages
7	Account created	115,192	115,192	0	0.00%	44.30%	Technical completion

Step	Stage	Customers entering	Completed	Abandoned	Stage abandonment	Cumulative reach	Current interpretation
							point in the available event data

Activation baseline

Metric	Current value
Total sign-ups	260,000
Completed onboarding	115,192 / 260,000 = 44.30%
Activated within 30 days	50,024 / 260,000 = 19.24%
Activation among onboarding completers	50,024 / 115,192 = 43.43%
Completed onboarding but not activated	65,168 customers
Share of completers not activated	56.57%

Selfie/liveness measurement note

Two metrics are useful, but they answer different questions:

- **Canonical funnel metric:** 34.16% customer-level stage abandonment, calculated using distinct customers who reach selfie/liveness and do not move past it.
- **Operational retry metric:** 22.28% attempt-level abandonment across 292,307 selfie attempt rows, with a 34.78% retry rate.

The customer-level metric is the metric used for product prioritization. The attempt-level metric is used to monitor retry and camera-flow friction. A row is not a person.

Key segment signals

Segment	Volume / share	Completion or abandonment signal	Activation signal	Product implication
Paid social	34.0% of base	29.81% complete	8.73% activate	Largest low-conversion opportunity

Segment	Volume / share	Completion or abandonment signal	Activation signal	Product implication
Organic	26.8% of base	52.20% complete	24.67% activate	Higher-intent comparison group
Referral	15.1% of base	65.01% complete	38.25% activate	Strong intent signal; avoid unnecessary spend
Paid search	16.1% of base	45.85% complete	17.87% activate	Mid-performing acquisition segment
Affiliate	8.1% of base	37.31% complete	12.66% activate	Lower-performing segment
Android	71.9% of base	42.16% complete	18.39% activate	Primary device context for selfie intervention
iOS	28.1% of base	49.80% complete	21.42% activate	Comparison group; no equivalent version effect
Current app version, 4.2.0+	61.3% of base	49.92% complete	21.61% activate	Reference group for version comparison
Legacy app version, 3.9–4.1.1	38.7% of base	35.42% complete	15.49% activate	Technical-risk population
Prior bank relationship	41.0% of base	47.13% complete	22.47% activate	Positive propensity signal
No prior bank relationship	59.0% of base	Lower than prior-relationship group	Lower than prior-relationship group	Greater need for education and guidance

At liveness, paid social abandonment is 47.37%, compared with 19.62% for referral. The highest-priority technical segment is paid social + Android + legacy app version: 19,596 customers reach liveness, 68.47% abandon and 13,417 are lost. The comparison group, paid social + Android + current app version, has 21,416 liveness entrants and 36.16% abandonment.

Across all channels, Android legacy versus current versions show 49.75% versus 25.97% liveness abandonment, a 23.8 percentage-point gap. The equivalent iOS comparison is 28.23% versus 25.46%, a 2.8 percentage-point gap. This supports an Android-specific intervention rather than a generic legacy-version prompt across all devices.

Document upload is a secondary opportunity: uploads taking more than 180 seconds have 23.37% abandonment compared with 12.25% for faster uploads.

Solution space

The team should bring multiple alternatives to the hackathon before committing to the MVP. The assessment below uses relative impact and cost because Engineering effort, channel cost, operational cost and incentive budget are not yet available.

#	Alternative	Problem addressed	Target segment	Role of AI	Relative impact	Relative cost	Speed	Key dependency	Decision
1	Version-aware update nudge before selfie	Customers are exposed to liveness friction on legacy app versions	Android only , legacy app, especially paid social — iOS excluded (see §3.2, effect is not present on iOS)	Deterministic eligibility; AI is optional for copy personalization	High ▾	Low ▾	Fast ▾	App-version mapping, update deep link, event tracking	Select for MVP
2	Adaptive selfie capture coach	Customers retry without understanding how to improve capture quality	Customers with retry or liveness failure signals	Classify friction archetype and select approved guidance	High ▾	Med... ▾	Med... ▾	UX/design, camera signals, approved guidance	P1 / extension
3	Assisted-verification fallback	Customers remain blocked after repeated failures	Customers with 2–3 failed attempts	Route and summarize the case; no automated identity decision	Med... ▾	High ▾	Slow ▾	Operations, compliance, capacity, service-level design	Future / pilot

#	Alternative	Problem addressed	Target segment	Role of AI	Relative impact	Relative cost	Speed	Key dependency	Decision
4	First-Use Activation Coach	Customers complete onboarding but do not make a first transaction	Completed onboarding, not yet activated, prioritized by incremental opportunity	Predict activation risk, select next-best action, generate constrained explanation	High ▾	Med... ▾	Med... ▾	Post-account-create events, approved action playbooks	Select for MVP
5	Acquisition-channel mix shift toward referral	Paid channels produce lower activation than referral	Acquisition strategy and new prospects	Optimize allocation only after causal and economic validation	Med... ▾ long term	High ▾	Slow ▾	CAC, LTV, scale, channel quality and attribution data	Strategic follow-up

Decision framework

The prioritization score should be calculated as:

Priority = expected incremental activations × confidence × actionability / implementation cost

The final decision must also satisfy two guardrails:

- Cost per incremental activated customer must remain within the business threshold defined by Product/Finance.
- Customers with high natural propensity to convert must not be unnecessarily interrupted, incentivized or contacted.

Strategic decision: fix the funnel versus change channel mix

The team will fix the onboarding funnel and the post-onboarding activation gap rather than make acquisition-channel reallocation the primary intervention.

Paid social is the largest channel at 34.0% of sign-ups and has the lowest activation rate at 8.73%, while referral activates at 38.25%. This makes channel mix an important strategic observation, but not the best two-day product intervention.

The decision is based on four principles:

- The selfie intervention improves the experience for customers across every acquisition channel, including paid social.
- The update nudge is narrowly targeted, low cost and directly connected to an observed Android/version signal.
- The activation coach addresses the second major loss that funnel repair alone cannot solve.
- Changing channel mix does not repair the product experience and requires a separate channel-economics strategy.

Channel mix optimization remains a future Growth strategy; it is not part of the MVP scope.

Recommended product

MVP solution A — Selfie Recovery: Version-Aware Update Nudge

Product concept

Before opening the selfie camera, the app detects whether the customer is on Android and a legacy app version. If eligible, it shows a lightweight, dismissible update nudge that helps the customer update the app and return directly to selfie/liveness.

Target segment

- Primary audience: 89,845 customers, representing 34.6% of the base, on Android and app versions 3.9.0–4.1.1.
- Highest-risk subsegment: paid social + Android + legacy version customers reaching liveness, where 68.47% abandon.
- Excluded from the update nudge: iOS customers, because the equivalent legacy/current version gap is only 2.8 percentage points.

User flow

1. The customer reaches selfie/liveness.
2. The system reads device OS and app version.
3. If the customer is Android and on a legacy version, the nudge appears before camera launch.
4. The customer can update the app or continue without updating.
5. After returning to the app, the customer resumes directly at selfie/liveness.
6. If the customer retries, the same prompt is suppressed according to frequency rules.
7. Exposure, update click, return and selfie completion are recorded.

Example copy:

Update the app. Update to the latest version for a smoother identity check.

Requirements

- Eligibility must be deterministic and auditable.
- The intervention must apply only to Android legacy versions in the defined range.
- The nudge must never permanently block onboarding.
- The customer must be able to continue without updating.
- The customer must return to the same selfie/liveness step after updating.
- The prompt must be frequency-capped.
- The prompt must support localized, concise and accessible copy.
- The original journey must remain available if the update is impossible or declined.

AI Role

AI is not required to decide eligibility. AI supports the product by:

- generating concise copy variants within an approved template;
- adapting language and tone to the customer's locale;
- explaining the decision in internal logs using only observable signals;
- summarizing experiment results.

The eligibility policy remains deterministic so the intervention is explainable and easy to audit.

Impact and cost

The target segment's completion is expected to move from approximately 33.9% toward the 49.8% level observed for Android customers on current versions. This corresponds to approximately 14,000 additional onboarding completions for the full Android-legacy segment and approximately 5,000–6,000 additional activations when the current activation relationship is applied.

The intervention has low relative cost: client-side banner or card, no monetary incentive and no new operational queue. The most important costs are implementation, app-version mapping, update-link handling and event instrumentation.

MVP solution B — First-Use Activation Coach

Product concept

After account creation, an eligible customer who has not yet made a first transaction receives one focused, personalized next step instead of a generic home-screen experience or a list of competing possibilities.

The product uses a propensity score to decide how much intervention is appropriate, then selects one approved action and generates a short explanation or guided task.

Eligible population

- Customer has completed onboarding.
- Customer has not yet made a first transaction.
- Customer is within the first-use activation window.

- Customer has not already received the maximum allowed intervention.

The eligible population is 65,168 customers who completed onboarding but did not activate within 30 days.

Proposed action playbook

The initial action set should be validated by BA and Product. A practical MVP can start with three actions:

Customer context or signal	Recommended action	Example guidance
No first-use event and no clear preference	Add money / make an initial transfer	"Add money to your account to start using it."
Signals suggest payment use case	Pay a bill or make a Pix payment	"Try your first payment in a few simple steps."
Card-oriented intent or available card context	Use the card for a first purchase	"Make your first purchase and see your account in action."

The product must not claim to know the customer's motivation unless that signal exists. When signals are ambiguous, use a neutral recommendation or ask the customer to choose.

Propensity tiers

The model predicts the probability of activated_30d among customers who completed onboarding. It is a propensity model, not an uplift model.

Tier	Typical signals	Product action	Incentive	Rationale
High propensity	Referral, prior banking relationship, organic acquisition	Light reminder or no intervention	None	These customers may convert without help; avoid unnecessary contact and spend
Medium propensity	Mixed signals; no single strong positive or negative indicator	Guided first-transaction task with incentive	Pay-on-conversion	Best efficiency opportunity: persuadable customers with room to improve
Low propensity	Paid social, no referral, no prior banking relationship	Low-cost educational nudge	None	Test relevance cheaply before committing budget

The medium tier is expected to represent approximately 40% of completers, or roughly 46,000 customers.

Model and feature contract

The activation classifier uses only information available before the first transaction:

- acquisition channel;
- device operating system;
- app version;
- age band;
- state;
- previous banking relationship;
- referral status;
- time spent in onboarding steps;
- selfie retry count;
- slow-document-upload flag;
- other customer-level onboarding signals available at the decision timestamp.

The following fields are excluded because they are outcomes or direct derivatives of the outcome:

- *activated_30d* as an input feature;
- *first_transaction_ts*;
- *days_to_first_transaction*;
- any post-transaction event;
- any feature computed using information after the intervention decision.

A gradient-boosting classifier is used because it captures interactions such as Android × app version and handles the mixed feature set. The output is a calibrated propensity score used for tiering and prioritization.

Next-best-action playbook

The coach recommends one primary action from an approved set. It does not claim to know the customer’s motivation when no supporting signal exists.

Observable context	Recommended action	Example guidance
No first-use event and no clear preference	Add money or make an initial transfer	“Add money to your account and start using it.”
Signals suggest payment use	Pay a bill or make a Pix payment	“Try your first payment in a few simple steps.”
Card-oriented context or available card	Make a first purchase with the card	“Make your first purchase and see your account in action.”

Observable context	Recommended action	Example guidance
Ambiguous context	Neutral choice between approved actions	“Choose the first way you’d like to use your account.”

The customer can accept the recommended action, choose another approved action or dismiss the coach. Dismissal never restricts account access.

Incentive design

The medium-propensity treatment can include a small, time-bound incentive such as cashback on the first transaction within seven days.

The incentive is pay-on-conversion:

- no payout when the customer is merely exposed to the message;
- payout only when the customer completes the qualifying first transaction;
- no incentive for high-propensity customers;
- no incentive in the low-propensity educational-nudge treatment.

This design keeps intervention cost close to the incentive face value per converted customer and reduces spend on customers who would likely activate without assistance.

AI architecture

1. Eligibility layer determines whether the customer is completed-but-not-activated.
2. Propensity layer produces the activation score.
3. Tiering policy assigns high, medium or low propensity.
4. Policy engine selects whether to intervene and which action is allowed.
5. Generation layer produces a short message from approved content.
6. Experience layer displays the card or guided task.
7. Measurement layer records exposure, action start, action completion, first transaction and incentive payout.

The LLM operates inside constrained templates or a retrieval-based approved content set. It must not invent products, benefits, deadlines, fees, eligibility conditions or reasons for customer behavior.

Impact and cost

If the medium tier contains approximately 46,000 completers and the incentivized coach creates a 10 percentage-point activation lift, the planning impact is approximately 3,200–4,600 incremental activations per cohort, depending on the final eligible population and treatment allocation.

The cost profile is favorable because:

- high-propensity customers receive no incentive;
- low-propensity customers receive a low-cost nudge;
- the medium-tier incentive is paid only after conversion;
- the experiment measures incremental activation rather than gross activation.

Metrics and experiment design

North Star

`activated_30d` = customers with a first transaction within 30 days / total sign-ups

MVP A metrics

Primary metric:

- Selfie/liveness completion rate among eligible Android legacy customers.

Secondary metrics:

- onboarding completion;
- successful app-update return;
- selfie retry rate;
- `activated_30d`;
- time from update prompt to selfie completion.

Guardrails:

- prompt dismissal and abandonment;
- support contacts;
- app crashes or technical errors;
- repeated prompt exposure;
- completion among ineligible customers;
- negative effect on current-version Android and iOS comparison groups.

MVP B metrics

Primary metrics:

- first transaction within seven days among eligible customers;
- first transaction within 30 days among eligible customers;
- incremental activation by propensity tier.

Secondary metrics:

- coach exposure;
- action start;
- action completion;
- message dismissal;

- time to first transaction;
- incentive payout rate.

Guardrails:

- cost per incremental activated customer;
- activation among high-propensity holdout versus treated customers;
- complaints and opt-outs;
- support contacts;
- incentive abuse or duplicate qualification;
- cannibalization of customers who would have activated without treatment.

Experiment design

Each MVP uses randomized treatment and control within the eligible population.

Selfie Recovery:

- Control: current selfie/liveness experience.
- Treatment: Android legacy version update nudge before camera launch.

First-Use Activation Coach:

- High propensity: current experience or light reminder, with a holdout to measure natural activation.
- Medium propensity: control versus guided first-transaction coach with pay-on-conversion incentive.
- Low propensity: control versus educational nudge without incentive.

The activation strategy is a heuristic v1. Propensity estimates who is likely to activate; they do not estimate the causal effect of an intervention. The embedded A/B test creates the treatment data needed for a future uplift model.

Risks and guardrails

Risk	Mitigation
Correlation is mistaken for root cause	Use the Android/version pattern for targeting, but measure causal impact with randomized treatment
LLM invents the reason for a failed selfie	Use approved templates and neutral fallback copy
High-intent customers are unnecessarily interrupted	Suppress high-propensity incentives, cap reminders and maintain holdouts

Risk	Mitigation
Update nudge blocks customers who cannot update immediately	Make it dismissible and preserve the original path
Incentive cost exceeds incremental value	Use pay-on-conversion and limit incentive eligibility to the medium tier
Incentive abuse or duplicate qualification	Enforce one qualifying transaction, one incentive and explicit eligibility rules
Model leakage inflates performance	Score only with information available at the decision timestamp; exclude transaction-outcome fields
Generic first-use action is not relevant	Use a small approved playbook and select one action from observable context
Intervention increases support demand	Track support contacts, cap frequency and use clear, concise copy
Channel-mix recommendation distracts from the MVP	Keep acquisition optimization as a separate strategic follow-up

Expected business impact

The two MVPs address different parts of the same North Star equation.

Selfie Recovery

- Addresses the largest onboarding-stage loss.
- Targets Android legacy customers, especially paid social customers at liveness.
- Has low relative cost and no monetary incentive.
- Planning impact: approximately 14,000 additional onboarding completions in the full Android-legacy segment and approximately 5,000–6,000 additional activations if the version gap is substantially reduced.

First-Use Activation Coach

- Addresses the 65,168 customers who complete onboarding but do not activate.
- Uses propensity tiers to avoid treating all 115,192 completers the same.
- Concentrates paid intervention in the approximately 46,000-customer medium tier.
- Planning impact: approximately 3,200–4,600 incremental activations if the medium-tier treatment creates a 10 percentage-point lift.
- Cost control: incentive cost is incurred only after the qualifying first transaction.

Together, the MVPs improve both multiplicative factors in the customer journey: the probability of completing onboarding and the probability of reaching first use after account creation.

Hackathon demo narrative

1. Start with 260,000 sign-ups and the seven-step funnel.
2. Show 44.30% onboarding completion and 19.24% activation within 30 days.
3. Highlight selfie/liveness as the largest stage loss: 65,129 customers and 44.98% of all observed abandonment.
4. Show the Android legacy versus current comparison and the absence of the same effect on iOS.
5. Demonstrate the version-aware update nudge before camera launch.
6. Show the event trail: exposure, update click, return and selfie completion.
7. Move to a completed-but-not-activated customer.
8. Show the propensity tier and why intervention intensity differs.
9. Demonstrate the medium-tier guided first transaction with a constrained AI-generated message.
10. Show that the incentive is pay-on-conversion, while high-propensity customers are protected from unnecessary spend.
11. Close with control versus treatment metrics and the combined North Star impact.

Final recommendation

Build and demonstrate two complementary MVPs:

1. Selfie Recovery — Version-Aware Update Nudge for Android customers on legacy app versions.
2. First-Use Activation Coach using propensity tiers, next-best action and pay-on-conversion incentives for the medium-propensity segment.

This is the strongest product decision because it combines:

- the highest-leverage funnel repair;
- a low-cost, technically specific intervention;
- an AI-native personalization layer;
- a direct connection to first transaction;
- controlled incentive economics;
- protection for customers who would likely convert without help.

The key message is:

We do not treat every customer the same. We remove the most actionable technical friction before selfie verification, then use propensity-based guidance to

help the right customers reach first value — while avoiding unnecessary spend on customers who would convert anyway.